IGCSE 0607 Practice Questions — E4.5

Parallel lines

Total Questions: 18 — **Total Marks:** 110 Marks

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  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

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1 Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

2 Line L has the equation $4x + 2y = 9$.
Find the equation of the line parallel to L that passes through the point $(3, -1)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L has the equation $y = 6 - 3x$.

Find the equation of the line parallel to L that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 Line L_1 has the equation $y = 3x - 7$.

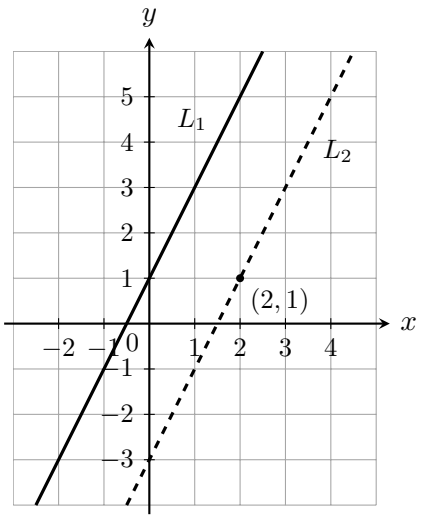
(a) Write down the gradient of a line that is parallel to L_1.

Gradient = ... **[1]**

(b) Line L_2 is parallel to L_1 and passes through the point $(2, -5)$.
Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

5 The diagram shows two parallel straight lines, L_1 and L_2, on a coordinate grid.
Line L_1 has equation $y = 2x + 1$ and line L_2 passes through the point $(2, 1)$.



(a) Write down the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2 in the form $y = mx + c$.

$y =$ ..... **[2]**

(c) A third line L_3 is parallel to L_1 and passes through the point $(-1, 5)$.
Find the equation of line L_3 in the form $y = mx + c$.

$y =$ ..... **[2]**

6 A straight line L_1 has the equation $2y - 5x = 8$.

(a) Find the gradient of line L_1.

..... **[2]**

(b) Line L_2 is parallel to L_1 and passes through the point $(0, -3)$.
Write down the equation of L_2.

$y =$ ..... **[2]**

(c) Line L_3 is parallel to L_1 and passes through the point $(4, 1)$.
Find the equation of L_3 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

7 The points A and B have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line L is parallel to the line segment AB and passes through the point $(4, -2)$.

(a) Find the gradient of the line segment AB.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

8 Line A has the equation $3x + 2y = 14$.

(a) Find the gradient of line A.

Gradient = ... **[2]**

(b) Line B is parallel to line A and passes through the point $(-4, 9)$.
Find the equation of line B.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[4]**

9 Line L_1 has equation $3x + 4y = 12$.

(a) Find the gradient of line L_1.

..... **[2]**

(b) Line L_2 is parallel to line L_1 and passes through the point $(4, 3)$.
Find the equation of line L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(c) Line L_2 crosses the x-axis at A and the y-axis at B.
Find the coordinates of the midpoint of AB.

(..... , ...) **[3]**

10 Line P has equation $4x - ky = 8$.
Line Q has equation $y = (k - 3)x + 2$.
The lines P and Q are parallel.

(a) Find the two possible values of the constant k.

$k =$ ..... **[4]**

(b) Given that line P has a positive gradient,

(i) find the equation of line P in the form $y = mx + c$.

$y =$ ..... **[2]**

(ii) A line R is parallel to line P and passes through the point $(-5, 7)$.
Find the equation of line R in the form $ax + by = c$, where a, b, and c are integers.

..... **[2]**

11 Line M has the equation $5x - 2y = 15$.
Point P has coordinates $(-4, 1)$.
Find the equation of the line parallel to M that passes through P.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

12 A is the point $(-3, 2)$ and B is the point $(5, 6)$.

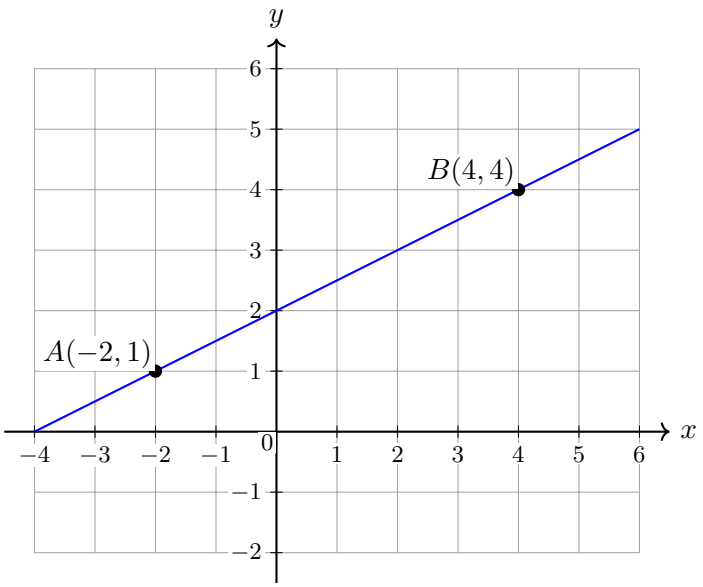
(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Line L is parallel to AB and passes through the point $(4, -1)$.
Find the equation of line L, giving your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

13 The diagram shows the points $A(-2, 1)$ and $B(4, 4)$ joined by a straight line.



(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Line M passes through the point $(2, -3)$ and is parallel to the line AB.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line M intersects the x-axis at point P and the y-axis at point Q.
Calculate the length of the line segment PQ.
Give your answer correct to 3 significant figures.

..... **[4]**

14 Points $A(-1, 5)$ and $B(3, -3)$ lie on line L_1.
Line L_2 is parallel to L_1 and passes through the point $C(2, 7)$.
Line L_2 crosses the x-axis at D and the y-axis at E.

(a) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) Calculate the exact area of triangle ODE, where O is the origin $(0, 0)$.

..... $units^2$ **[3]**

15 The points $P(-2, 3)$, $Q(4, 15)$ and $R(k, 2k + 1)$ lie on a coordinate plane, where k is a constant.

(a) Line L passes through R and is parallel to the line segment PQ.
Given that the y-intercept of line L is -5, find the value of k.

$k =$ ..... **[4]**

(b) Find the equation of the line passing through $Q(4, 15)$ that is parallel to the line $5x - 2y + 9 = 0$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

16 Line L_1 has the equation $3x - 4y = 12$.
Line L_2 is parallel to L_1.

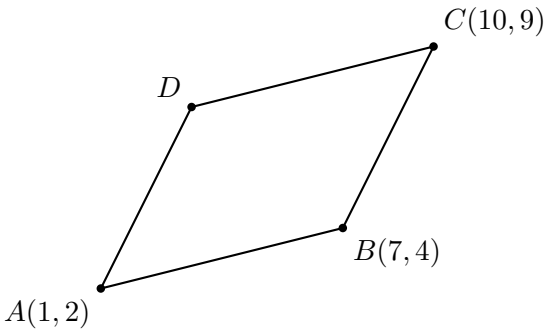
Line L_2 crosses the x-axis at point A and the y-axis at point B.
The length of the line segment AB is 15.

Find the two possible equations of L_2.
Give your answers in the form $y = mx + c$.

.....

..... **[7]**

17 The points $A(1, 2)$, $B(7, 4)$, and $C(10, 9)$ are three vertices of a parallelogram $ABCD$.
Line CD is parallel to line AB and line AD is parallel to line BC.



Not to scale

(a) Find the equation of the line CD.

..... **[3]**

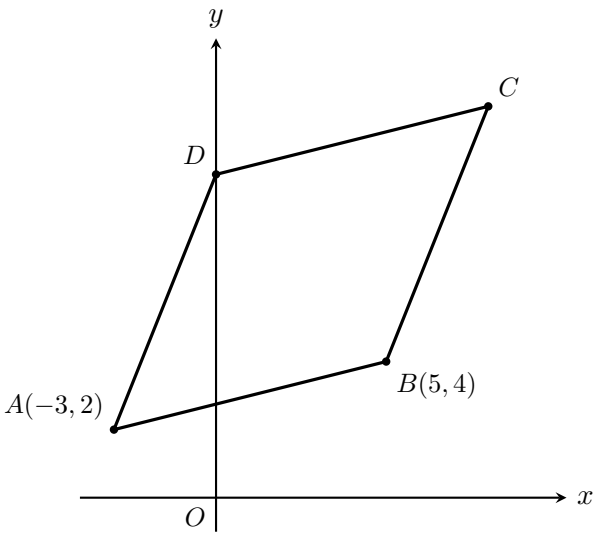
(b) Find the equation of the line AD.

..... **[3]**

(c) Work out the coordinates of the vertex D.

(..... , ...) **[3]**

18 The diagram shows a sketch of a parallelogram $ABCD$ on a coordinate grid.
Vertex A has coordinates $(-3, 2)$ and vertex B has coordinates $(5, 4)$.
The line passing through B and C has equation $5x - 2y = 17$.



(a) Find the equation of the line passing through A and D in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Vertex D lies on the y-axis.
Write down the coordinates of D and find the coordinates of vertex C.

$D =$ ..... , $C =$ ..... **[3]**

(c) Line L is parallel to AB and passes through the midpoint of BC.
Find the equation of line L in the form $ax + by = c$, where a, b, and c are integers.

..... **[4]**

(d) Calculate the length of diagonal AC, giving your answer correct to 3 significant figures.

$AC =$ ..... **[2]**